Assignment No: 3

Build Automation of a Web Application Using Jenkins

Aim

To configure Jenkins to retrieve the Student Task Manager application from GitHub and execute the build successfully.

Output

1. Local build verification

```
aditya@aditya:~/devops-task-manager$ npm install
added 57 packages, and audited 58 packages in 3s
found 0 vulnerabilities
aditya@aditya:~/devops-task-manager$ npm run build
> task-manager@1.0.0 build
> node build.js
Starting application build validation...
Application files validated successfully.
Build completed successfully.
```

2. Push to GitHub

```
aditya@aditya:~/devops-task-manager$ git add .
aditya@aditya:~/devops-task-manager$ git commit -m "Add application build configuration"
aditya@aditya:~/devops-task-manager$ git push origin main
Writing objects: 100% (4/4), done.
```

3. Jenkins job configuration

```
Job Name         : task-manager-build
Project Type     : Freestyle Project
SCM              : Git
Repository URL   : https://github.com/aditya/devops-task-manager.git
Branch           : */main
Build Step       : npm install
                   npm run build
```

4. Console Output - Successful Build (#1)

```
Started by user aditya
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/task-manager-build
> git fetch --tags --progress -- https://github.com/aditya/devops-task-manager.git
Checking out Revision 4b8e32a (refs/remotes/origin/main)
[task-manager-build] $ /bin/sh -xe /tmp/jenkins.sh
+ npm install
added 57 packages in 3s
+ npm run build
Starting application build validation...
Application files validated successfully.
Build completed successfully.
Finished: SUCCESS
```

5. Console Output - Failed Build (#2, public/index.html removed)

```
+ npm run build
Starting application build validation...
Build Failed: public/index.html not found
Build step marked build as failure.
Build step 'Execute shell' marked build as failure
Finished: FAILURE
```

6. Build history

```
#2   FAILURE   public/index.html removed
#1   SUCCESS   Initial build
```

Conclusion

Jenkins was integrated with GitHub, pulled the source, installed dependencies and executed the build. Both a successful and a failed build were analysed through the console output.